

# Influ-BERT: a domain-adaptive genomic language model for advancing influenza A virus research

Rongye Ye<sup>1,2,3,4,†</sup>, Lun Li<sup>1,2,3,†</sup>, Ana Tereza Ribeiro de Vasconcelos<sup>5,\*</sup>, Shuhui Song<sup>1,2,3,4,\*</sup>

<sup>1</sup>National Genomics Data Center, China National Center for Bioinformation, No. 1 Beichen West Road, Chaoyang District, Beijing 100101, China

<sup>2</sup>Beijing Key Laboratory of Intelligent Governance and Application of Biological Big Data, China National Center for Bioinformation, No. 1 Beichen West Road, Chaoyang District, Beijing 100101, China

<sup>3</sup>Beijing Institute of Genomics, Chinese Academy of Sciences, No. 1 Beichen West Road, Chaoyang District, Beijing 100101, China

<sup>4</sup>University of Chinese Academy of Sciences, No. 1 Yanqihu East Rd, Huairou District, Beijing 101408, China

<sup>5</sup>Bioinformatics Laboratory, National Laboratory for Scientific Computing, 333 Getúlio Vargas Avenue, Quitandinha, Petrópolis, RJ 25651-076, Brazil

\*Corresponding authors. Ana Tereza Ribeiro de Vasconcelos, National Laboratory for Scientific Computing, Bioinformatics Laboratory, Petrópolis 25651-075, Brazil. E-mail: atrv@lncc.br; Shuhui Song, National Genomics Data Center, China National Center for Bioinformation, Beijing 100101, China. E-mail: songshh@big.ac.cn.

<sup>†</sup>Rongye Ye and Lun Li contributed equally.

## Abstract

Influenza A virus (IAV) poses a persistent threat to global public health due to its broad host adaptability, frequent anti-genic variation, and potential for cross-species transmission. Accurate identification of IAV subtypes is essential for effective epidemic surveillance and precise disease control. Here, we present Infl-BERT, a domain-adaptive pretrained model based on the Transformer architecture. Optimized from DNABERT-2, Infl-BERT was developed using a dedicated corpus of ~900 000 influenza genome sequences. We constructed a custom Byte Pair Encoding tokenizer, and employed a two-stage training strategy involving domain-adaptive pretraining followed by task-specific fine-tuning. This approach significantly enhanced identification performance for IAV subtypes. Experimental results demonstrate that Infl-BERT outperforms both traditional machine learning approaches and general genomic language models, such as DNABERT-2, Nucleotide Transformer, and MegaDNA, in the task of IAV subtype identification. The model consistently achieved F1-scores above 97% across five subtype classification tasks and exhibited stable performance gains for subtypes that are underrepresented in sequencing data, including H5N8, H1N2, and H13N6. Beyond subtype identification, Infl-BERT was successfully applied to additional tasks including respiratory virus identification, IAV pathogenicity prediction, and identification of IAV genomic fragments and functional genes, demonstrating robust performance throughout. Further interpretability analysis using sliding window perturbation confirmed that the model focuses on biologically significant genomic regions, providing insight into its improved predictive capability.

**Keywords** IAV subtype identification, IAV Pathogenicity Prediction, genomic language model, Infl-BERT, influenza virus

## Introduction

Influenza viruses are among the most epidemiologically significant respiratory pathogens globally, posing a long-standing and significant threat to global public health [1, 2]. Among these, influenza A viruses (IAV) constitute the primary pathogen responsible for human influenza pandemics due to its broad host range and capacity for antigenic variation. IAV is an RNA virus, and its genome is composed of eight single-stranded negative-sense RNA segments [3]. Specifically, Segment 4 encodes hemagglutinin (HA), and Segment 6 encodes neuraminidase (NA). These two surface glycoproteins determine viral subtype characteristics; they mediate viral entry into and release from host cells and are the primary targets for the host immune system. The generation of new HA-NA combinations through gene reassortment provides the molecular basis for cross-

species transmission and pandemics by IAV [4]. For instance, the highly pathogenic avian influenza virus of the H5N1 subtype is one of the most significant pandemic threats currently identified. They have demonstrated the capacity for transmission among multiple mammalian species, indicating a potential for cross-species transmission to humans [5]. The widespread prevalence of H9N2 in live bird markets increases the zoonotic risk [6]. Similarly, H7N9, an avian influenza virus that can be highly pathogenic in poultry, poses a considerable threat to the poultry industry and public health, with human infections associated with a high mortality rate [7].

Traditional IAV subtype detection mainly relies on molecular biology techniques like reverse transcription polymerase chain reaction (RT-PCR) for accurate identification of common subtypes

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[8, 9]. Serological methods, including hemagglutination inhibition (HAI) assays and neutralization assays, are also widely employed. HAI detects subtype-specific antibodies by their ability to inhibit hemagglutination [10], while neutralization assays identify subtypes by measuring antibody-mediated neutralization of viral infectivity [11]. However, these conventional methods have limited throughput, posing challenges for large-scale automated sample analysis. Advances in high-throughput sequencing have enabled genome-based methods for influenza subtyping, primarily through sequence alignment against reference databases [12]. These approaches recognize subtypes based on sequence similarity and facilitate the analysis of evolutionary relationships through phylogenetic trees [13, 14]. Alignment-based tools such as FluSurver and NextClade [15] leverage reference databases to perform subtype or clade assignment, and serve as essential benchmarks for frontline surveillance personnel. Despite their utility, these methods encounter inherent limitations in complex scenarios. A primary constraint stems from their dependence on pre-defined reference libraries, which restricts generalization to underrepresented or low-frequency subtypes. For example, we evaluated the performance of multiple tools across various low-frequency subtypes, with the results presented in [Supplementary Table S1](#). Furthermore, traditional alignment-based approaches rely primarily on HA/NA segments, limiting identification when these segments are absent in partial sequencing data.

The advent of artificial intelligence has transformed influenza subtype identification, with machine learning methods increasingly applied for this purpose. Current approaches typically use k-mer frequencies as features to train identification models [16–18]. However, the performance of these methods is highly dependent on manually designed features, such as the selection of k-mer length, making them sensitive to the quality feature extraction. Furthermore, these conventional models are limited in their ability to capture contextual dependencies within viral genome sequences. Recent advances in Transformer architecture have created new opportunities for biological sequence analysis [19–22]. Leveraging the symbolic similarity between nucleic acid sequences and natural language, researchers have developed numerous pretrained genomic language models [23, 24], such as DNABERT [25, 26], MegaDNA [27], and RNA-specific models like LucaProt [28], which integrating RdRp amino acid sequences with predicted protein structures in a multimodal Transformer framework, thereby enhancing virus discovery and the exploration of viral “dark matter.”

However, long-tail data distribution is prevalent across various real-world applications. This distribution pattern is characterized by a severe class imbalance, where most samples are concentrated in a small number of categories, while a large number of remaining categories contain relatively few samples. In the context of IAV, subtypes that have demonstrated sustained and widespread transmission among humans (e.g. H1N1 and H3N2) are classified as frequent subtypes. In our dataset, subtypes with an occurrence frequency below 15% and only limited documented human infection cases were classified as low-frequency subtypes. Due to the long-tail distribution of IAV data, models exhibit strong identification capabilities for prevalent subtypes (H1N1, H3N2), but performance declined significantly for extremely low-frequency subtypes. These low-frequency subtypes remain highly important for epidemiological surveillance [29–31]. Consequently, the development of robust subtype prediction models is highly valuable for early warning and prevention for influenza pandemics.

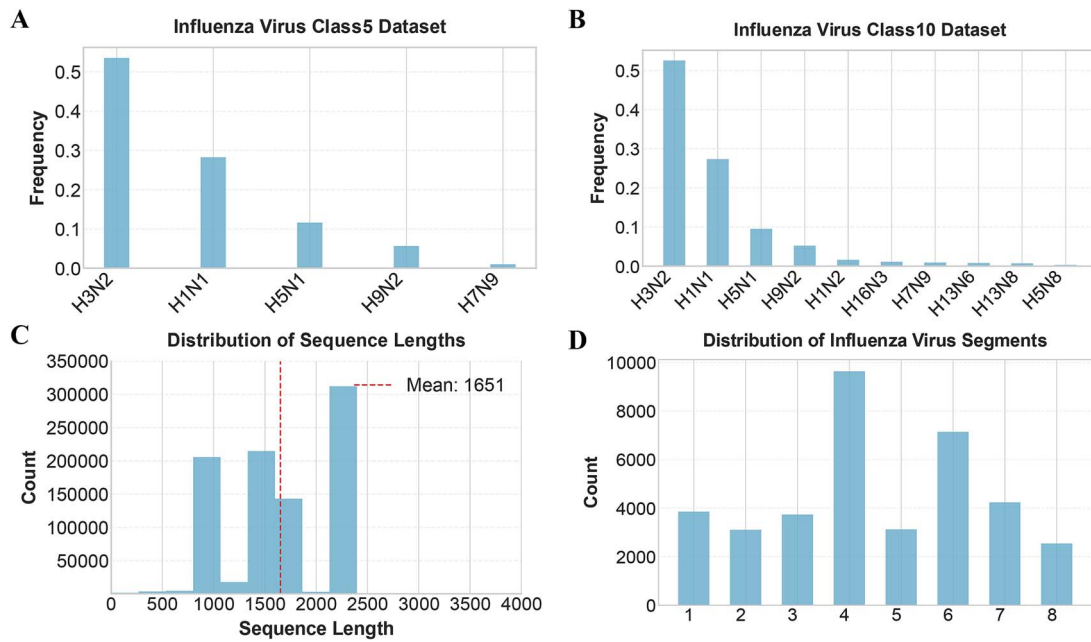
Here, we propose Infl-BERT, a model inspired by DNABERT-2 architecture, that was specifically optimized to enhance identification capabilities for low-frequency IAV subtypes. We evaluated Infl-BERT against several baseline approaches, including k-mer-based machine learning methods (random forest, RF; support vector machine, SVM; logistic regression, LR), masked pretrained models DNABERT-2 and Nucleotide Transformer, and the generative model MegaDNA. The results demonstrate that Infl-BERT outperforms these existing methods in identification low-frequency subtypes while maintaining strong overall performance. Moreover, to extend the applicability of Infl-BERT, we additionally designed three biologically meaningful downstream tasks: identification of common respiratory viruses (including SARS-CoV-2, rhinovirus, respiratory syncytial virus, and influenza virus), identification of influenza virus genomic fragments and functional genes, and prediction of IAV pathogenicity. Experimental results indicate that Infl-BERT achieves superior performance across these tasks, demonstrating its broad utility in influenza genome analysis. Finally, we introduced sequence perturbation analysis based on a sliding window approach, which revealed differences in feature identification patterns between Infl-BERT and DNABERT-2, providing a preliminary explanation for the performance improvement of Infl-BERT.

## Materials and methods

### Dataset

We systematically collected ~900 000 influenza virus genome sequences from the NCBI Nucleotide database [32], encompassing more than 100 naturally occurring HA–NA combinations that have been detected to date. The dataset comprises ~800 000 IAV sequences, ~100 000 influenza B virus sequences, and over 4000 influenza C and D virus sequences. This dataset was utilized in two key steps: (i) the construction of a Byte Pair Encoding (BPE) vocabulary specific to influenza virus sequences; and (ii) domain-adaptive pretraining of the language model. To evaluate the performance of the adapted model comprehensively, we designed four downstream tasks with biological relevance.

Task1 focuses on subtype identification. We first established an identification workflow that is agnostic to specific genomic fragments, encompassing tasks for 5 and 10 subtypes. This workflow is particularly suited for rapid detection scenarios and cases with limited sequencing coverage. Considering the requirements of high-precision applications, such as vaccine design, which necessitate the analysis of specific influenza genomic segments, we further developed a dedicated training and subtyping pipeline focused exclusively on HA and NA fragments. Notably, both scenarios adopt a single genomic segment as input, rather than concatenating multiple segments into a full-length genome sequence for modeling. For this task, to ensure rigorous evaluation and prevent data leakage, we implemented a robust isolation between the pretraining corpus and downstream datasets. Initially, we established an independent candidate pool comprising 200 000 IAV sequences, explicitly verifying zero overlap with the domain-adaptive pretraining corpus. We then performed internal sequence deduplication within this independent pool. To mitigate potential data leakage caused by highly homologous sequences, we employed MMseqs2 for homology-based clustering (thresholds: 98% sequence identity, 80% coverage). The 98% identity threshold was strategically selected to eliminate redundant near-duplicates while



**Figure 1** Statistical distribution of data. (a and b) the frequency distribution in the downstream task dataset; (c) length distribution of segments within the corpus used during the domain-adaptive pretraining phase; (d) distribution of the number of different genomic segments across downstream task datasets.

preserving biologically significant micro-variants driven by high-frequency mutations. Subsequently, we constructed distinct datasets for the five-class and 10-class tasks by sampling 40 000 sequences per task from this processed pool. The dataset exhibits a long-tail distribution (Fig. 1a and b), and a length distribution of sequences within the domain-adaptive pretraining corpus is presented in Fig. 1c. Crucially, dataset partitioning (training/validation/test ratio of 8:0.5:1.5) was conducted at the cluster level. This cluster-wise splitting strategy ensures that highly homologous sequences do not span across training and testing sets. The distribution of different genomic fragments is shown in Fig. 1d.

Task2 is designed to identify common respiratory viruses, including SARS-CoV-2, rhinovirus, respiratory syncytial virus (RSV), and influenza virus. Approximately 10 000 sequences were selected for each virus. For every virus type, the corresponding genomic fragments were extracted. To mitigate potential biases arising from variations in sequence length, all viral fragments were constrained to a maximum length of 10 000 base pairs. In addition, rigorous deduplication was performed to ensure that no duplicate sequences were present in the dataset.

Task3 focuses on the identification of IAV genomic fragments and functional genes. For each accession in the Task1 dataset, the corresponding fragment labels, functional gene sequences, and their associated annotations were retrieved from the NCBI database.

Task4 is dedicated to predicting the pathogenicity of IAV sequences. We constructed a label mapping dictionary to define pathogenicity categories (see Supplementary Table S2) and assembled a dataset of 1662 sequences with relatively well-defined pathogenicity categories. During dataset construction, internal sequence deduplication was performed to prevent redundancy from biasing model evaluation. To address the common challenge of limited data availability in real-world scenarios, this task specifically evaluates model performance under data-scarce conditions. Accordingly, we designed two limited-data learning experiments to systematically evaluate

predictive performance: one trained on 200 samples (100 high- and 100 low-pathogenicity) and the other on 400 samples (200 high- and 200 low-pathogenicity), with the remaining sequences reserved for independent testing.

With the exception of Task 2 and 4, all datasets were partitioned into training, validation, and test sets at the cluster level using a ratio of 8:0.5:1.5. Model performance was systematically evaluated using accuracy, precision, recall, and F1-score as primary metrics to provide a robust quantitative assessment.

## Experiment setting

### Software configuration

Given the relatively low sequence complexity of influenza viruses compared to the broad cross-species training data of DNABERT-2, we hypothesized that the original vocabulary size of 4096 might result in sparse representations. To empirically determine the optimal token granularity, we conducted a parameter ablation study (As shown in Table 1, Supplementary Fig. S1). Consequently, guided by the experimental results, we constructed an influenza-specific BPE vocabulary with a size of 1024 during the domain-adaptive pretraining phase. The new vocabulary reinitializes the embedding layer, while the Transformer Encoder weights are initialized from DNABERT2, preserving the backbone architecture. The parameter settings of InflBERT were kept the same as those of DNABERT-2 (As shown in Supplementary Table S3). All experiments were conducted using the PyTorch framework (version 2.5.1), with model training and evaluation implemented via the Hugging Face Transformers library (version 4.49.0).

### Baseline model

This experiment compares six models, including three machine learning models based on k-mer encoding, for which 3–6 mer representations were systematically evaluated and the best-performing results

**Table 1** Parameter ablation study of Infl-BERT.

| Influ-BPE  | Vocab_Size  | Use_Corpus | F1-score     |
|------------|-------------|------------|--------------|
| No         | 4096        | No         | 0.855        |
| No         | 4096        | Yes        | 0.889        |
| Yes        | 512         | Yes        | 0.902        |
| <b>Yes</b> | <b>1024</b> | <b>Yes</b> | <b>0.913</b> |
| Yes        | 2048        | Yes        | 0.898        |
| Yes        | 4096        | Yes        | 0.899        |

The results were tested on the IAV Subtype 10 task. Infl-BPE: "Yes" indicates that a newly trained vocabulary is used, and Use\_Corpus: "Yes" indicates that a domain-adaptive corpus is employed. The bold values indicate the best performance relative to the other models.

were reported (as shown in [Supplementary Table S4](#)), as well as DNABERT-2 and the Nucleotide Transformer, both based on masked language pre-training, and MegaDNA, which is based on generative pre-training. (i) **RF** [33]: an ensemble learning model that improves model generalization by combining the predictions of multiple decision trees. It is widely used in classification and regression tasks; (ii) **SVM** [34]: a supervised discriminative model whose core goal is to find an optimal hyperplane that maximizes the separation between samples of different classes; (iii) **LR** [35]: a commonly used classification machine learning model for classification that outputs a probability representing the likelihood of a sample belonging to a certain class. It is widely applied in classification tasks; (iv) **DNABERT-2** [26]: a self-supervised pretrained DNA language model that uses BPE encoding. It was pretrained on a large-scale multi-genome dataset containing 32.5 billion nucleotide bases and covering 135 species. This model can capture complex dependencies in DNA sequences and has demonstrated superior performance in multiple biological tasks; (v) **MegaDNA** [27]: a generative pretrained long-sequence DNA model that uses nucleotide-level tokenization. This model can handle genomic sequences up to 96 000 base pairs and has the capability for sequence generation and classification of unannotated sequences. (vi) **Nucleotide transformer (100 M)** [36]: the model integrates genomic information from 3202 human genomes and 850 genomes from diverse species, enabling the generation of context-specific representations of nucleotide sequences and achieving high-accuracy predictions even in low-data settings.

### Mask language model

Inspired by the success of natural language processing, masked pre-training has emerged as an effective paradigm for learning representations from genomic sequences. As a classical self-supervised learning strategy, masked pretraining operates by randomly masking a subset of input tokens during training and requiring the model to predict the original tokens based on their surrounding context. This approach enables the model to capture deep semantic representations of sequences. The objective function is designed to maximize the likelihood of the masked tokens, which is formally defined as follows:

$$\mathcal{L}_{lm} = -\sum_{i \in \mathcal{M}} \log P(x_i | X^{\text{mask}}), \quad (1)$$

where  $x_i$  represents the token at position  $i$  in the original sequence, and  $X^{\text{mask}}$  refers to the masked input sequence.  $P(x_i | X^{\text{mask}})$  denotes the model's probability of reconstructing  $x_i$  given the masked input  $X^{\text{mask}}$ . The Infl-BERT model adopts the same architectural framework as DNABERT-2, utilizing the Transformer attention mechanism

as its core component for sequence representation learning. The attention function [14] is computed as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (2)$$

where  $Q$  denotes the query matrix, representing the current input query;  $K$  denotes the key matrix, which typically encodes the features of candidate elements;  $V$  is the value matrix, containing the information associated with each key; and  $d_k$  is a scaling factor used to normalize the dot product of queries and keys. Within the Transformer architecture, knowledge representations across different subspaces are typically learned using a multi-head attention mechanism. The computation is defined as follows:

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \dots, \text{head}_h) W_O, \\ \text{head}_j &= \text{Attention}(QW_Q^j, KW_K^j, VW_V^j), \end{aligned} \quad (3)$$

where  $W$  denotes the learnable projection matrices applied during linear transformations, and  $\text{head}_j$  refers to the  $j$ -th attention head among a total of  $h$  heads in the multi-head attention mechanism. Finally, during the fine-tuning phase, the task-specific loss function is defined as the cross-entropy loss, which is formulated as follows:

$$\mathcal{L}_{CE} = -\sum_{i=1}^C y_i \log(p_i), \quad (4)$$

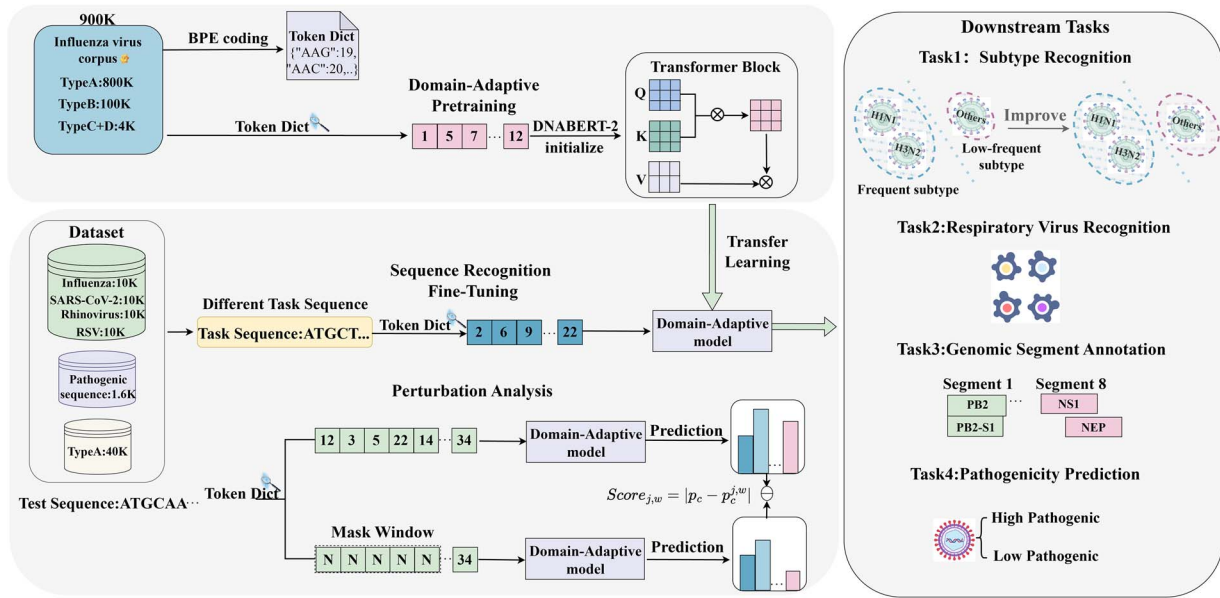
where  $C$  denotes the total number of classes,  $y_i$  represents the ground-truth label for the sample, and  $p_i$  denotes the predicted probability that the sample belongs to class  $i$ .

### Transfer knowledge via domain-adaptive continued pretraining

Although genome language models pretrained on cross-species genomic data have demonstrated strong performance across diverse biological tasks, their direct application to influenza virus genotyping is challenged by substantial distributional shifts. The inherently high mutation rates of influenza viruses, coupled with their long-tailed subtype distribution, create a semantic gap between the source and target domains, thereby limiting the fine-tuning effectiveness of general-purpose models such as DNABERT-2. To bridge this domain discrepancy, we propose a domain-adaptive continual pretraining strategy, aimed at aligning the model's representational space with the distributional characteristics of influenza viruses through self-supervised learning on target-domain corpora, thus providing a more suitable representation foundation for downstream tasks.

Specifically, we adopt the DNABERT-2 architecture as the backbone and introduce influenza-specific optimizations. First, we replace the general-purpose tokenizer with a virus-specific BPE tokenizer. Through parameter ablation, we determined an optimal vocabulary size of 1024, balancing token granularity and representational density. The embedding matrix of the input embedding layer is structurally adjusted to correspond to the vocabulary size and initialized from scratch to learn virus-specific token representations, while the Transformer encoder weights are initialized from the pretrained DNABERT-2 checkpoints. The model is then subjected to continual pretraining on our influenza corpus, where masked sequence reconstruction loss is minimized, effectively recalibrating the high-dimensional feature space to capture influenza-specific sequence dependencies.





**Figure 2** Schematic overview of the Influenza-BERT framework. A BPE tokenizer is trained on a curated influenza-specific corpus to construct a domain-specific vocabulary, which is subsequently used for domain-adaptive pretraining based on the DNABERT-2 architecture. The sequence recognition fine-tuning workflow is performed via transfer learning combined with task-specific fine-tuning after domain-adaptive pretraining. Interpretability analysis is conducted using a masking-based perturbation strategy to identify attention-sensitive regions, enabling interpretation of model predictions.

For downstream applications, task-specific classification heads are appended to the domain-adapted encoder, and full-parameter supervised fine-tuning is employed. This enables the model to retain generalizable genomic representations while simultaneously acquiring influenza-specific predictive capacity [37].

### Sequence perturbation analysis

The core idea of sequence perturbation analysis is to systematically mask local regions of the input sequence and then feed the masked sequences into the model to observe the resulting changes in output. This allows for a quantitative assessment of each sequence fragment's contribution to the model's decision. Specifically, for an influenza viral genome sequence  $S = (s_1, s_2, \dots, s_n)$ , and a set of class labels  $C = \{c_1, c_2, \dots, c_m\}$ , we define a window size  $w$ , and first compute the model's prediction on the original input sequence  $S$ :

$$P(S) = [p_1, p_2, \dots, p_m], \quad (5)$$

where  $\sum_{i=1}^m p_i = 1$ . Subsequently, for a window starting at position  $j$ , we set the  $w = 5$ , and apply masking to this region. The perturbed sequence is defined as:

$$\text{Mask}_j(S) = (s_1, \dots, s_{j-1}, N * w, s_{j+w}, \dots, s_n). \quad (6)$$

Here,  $N$  denotes the masking operation. Finally, for a sequence labeled as class  $c$ , the importance score of the window is computed as:

$$\text{Score}_{j,w} = |p_c - p_c^{j,w}|, \quad (7)$$

where  $p_c$  denotes the model-predicted probability of class  $c$  for the original input sequence, and  $p_c^{j,w}$  denotes the predicted probability of

class  $c$  after masking a window of size  $w$  starting at position  $j$  in the original sequence.

## Results

### Overview of Influenza-BERT

As shown in Fig. 2, the Influenza-BERT framework comprises by the domain-adaptive pretraining module and sequence recognition fine-tuning module. We first construct a vocabulary tailored for an influenza-specific corpus sequences using the BPE algorithm. Then, all influenza virus sequences in the corpus were tokenized using this vocabulary. The pre-trained model was initialized with the DNABERT-2 pretrained weights and conducted for a domain adaptive pretraining using masked language model objective. For different downstream tasks, we constructed three datasets and consistently applied the influenza-specific BPE vocabulary to tokenize the input sequences. Following a transfer learning paradigm, we leveraged the domain-adaptive model to extract features, upon which task-specific classification heads were added to enable end-to-end fine-tuning. Specifically, we designed four biologically meaningful downstream tasks: (i) improving the subtyping accuracy for low-frequency IAV subtypes; (ii) identifying common respiratory viruses; (iii) recognizing IAV genomic segments and functional genes; and (iv) predicting IAV pathogenicity. A sliding window-based perturbation analysis was further conducted to provide preliminary interpretability of the model's predictions.

### Performance evaluation for IAV subtype identification

We systematically evaluated the performance of Influenza-BERT on the task of IAV subtype identification without distinguishing genomic segments, encompassing high-frequency subtypes (H3N2 and H1N1) as well as low-frequency subtypes. This scenario is particularly relevant

**Table 2** Infl-BERT was compared with baseline models on subtype recognition.

| Task           | Model      | Accuracy     | Precision    | Recall       | F1_score     |
|----------------|------------|--------------|--------------|--------------|--------------|
| IAV subtype 5  | LR         | 0.967        | 0.968        | 0.967        | 0.966        |
|                | SVM        | 0.967        | 0.969        | 0.967        | 0.966        |
|                | RF         | 0.950        | 0.951        | 0.950        | 0.946        |
|                | DNABERT2   | 0.963        | 0.959        | 0.963        | 0.960        |
|                | MegaDNA    | 0.917        | 0.917        | 0.917        | 0.916        |
|                | NT         | 0.947        | 0.950        | 0.947        | 0.947        |
|                | Influ_BERT | <b>0.971</b> | <b>0.973</b> | <b>0.971</b> | <b>0.972</b> |
| IAV subtype 10 | LR         | 0.891        | 0.883        | 0.891        | 0.877        |
|                | SVM        | 0.846        | 0.841        | 0.846        | 0.828        |
|                | RF         | 0.871        | 0.849        | 0.871        | 0.849        |
|                | DNABERT2   | 0.873        | 0.860        | 0.873        | 0.855        |
|                | MegaDNA    | 0.841        | 0.826        | 0.841        | 0.831        |
|                | NT         | 0.892        | 0.889        | 0.892        | 0.887        |
|                | Influ_BERT | <b>0.917</b> | <b>0.912</b> | <b>0.917</b> | <b>0.913</b> |
| H/N subtype 10 | LR         | 0.971        | 0.952        | 0.971        | 0.959        |
|                | SVM        | 0.989        | 0.988        | 0.989        | 0.987        |
|                | RF         | 0.977        | 0.980        | 0.977        | 0.971        |
|                | DNABERT2   | 0.992        | 0.992        | 0.992        | 0.991        |
|                | NT         | 0.991        | 0.989        | 0.991        | 0.990        |
|                | MegaDNA    | 0.960        | 0.943        | 0.960        | 0.950        |
|                | Influ_BERT | <b>0.995</b> | <b>0.995</b> | <b>0.995</b> | <b>0.995</b> |

“IAV Subtype 5” and “IAV Subtype 10” refer to recognition tasks involving 5 and 10 IAV subtypes, respectively, without distinguishing specific genomic segments; whereas “H/N Subtype 10” denotes the recognition of 10 IAV subtypes based solely on the HA and NA segments. The bold values indicate the best performance relative to the other models.

for rapid detection and situations with insufficient sequencing coverage. Among these low-frequency subtypes, H5N1, H7N9, and H9N2, with frequencies 11.5%, 0.9%, and 5%, were prioritized for analysis due to their relatively significant impact on public health as reported in previous studies. We assessed performance using four metrics—accuracy, precision, recall, and F1-score, and found that Infl-BERT outperformed all other models across overall dataset performance metrics (As shown in Table 2).

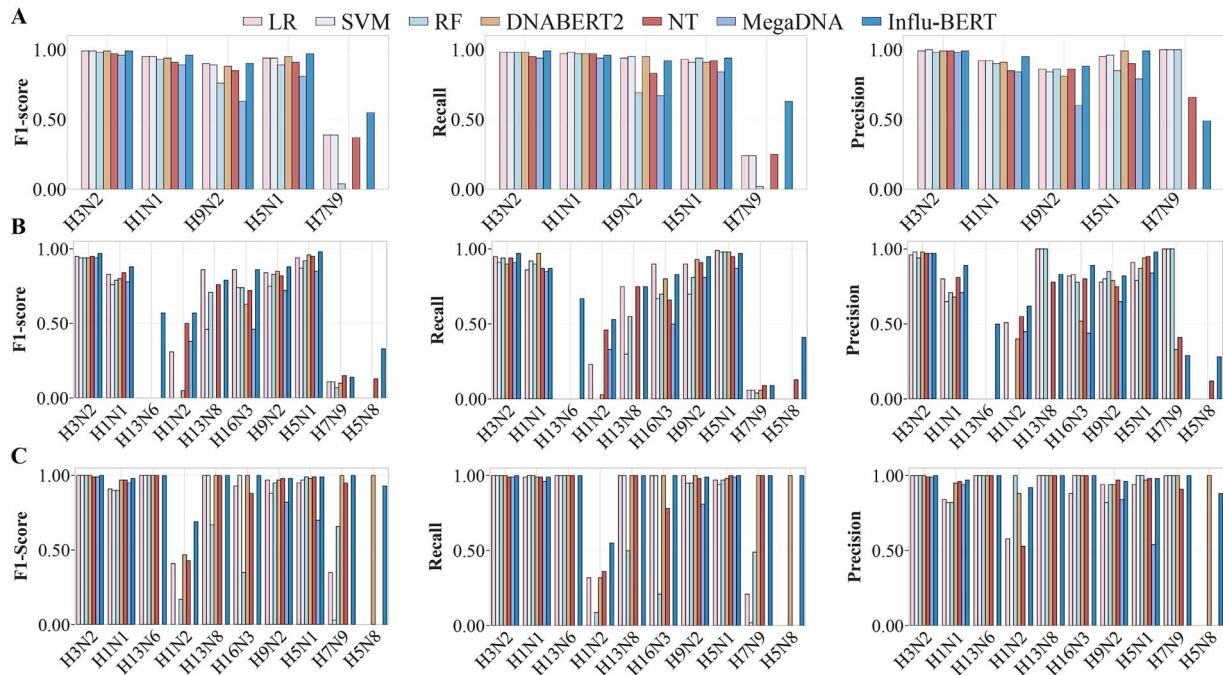
In terms of F1-score Fig. 3a and b, Infl-BERT demonstrated a significant advantage compared with the other five models. In the five-class(five-subtype) identification task, Infl-BERT achieved F1-scores of 97% for H5N1 and 90% for H9N2, and outperformed the second-best logistic regression model on H7N9 by 16%, demonstrating its robust performance across low-frequency IAV subtypes. In the 10-class (10-subtype) task, Infl-BERT outperformed other models on 8 of the 10 subtypes, especially for H13N6, H1N2, H5N8. Notably, in the H13N6 subtype identification task, most other models failed to recognize this subtype, while Infl-BERT achieved an F1-score of 57%, providing strong evidence for the effectiveness of Infl-BERT. Furthermore, Infl-BERT demonstrated consistent performance between recall and F1-score metrics, while demonstrating superior performance in identifying the majority of low-frequency subtypes compared with other benchmark models. Detailed results are provided in Supplementary Table S5–S13 and Supplementary Note S1.

In terms of precision, Infl-BERT demonstrated advantages over other models for the H1N1, H9N2 subtypes in the five-class classification task, and outperformed other models in six subtypes in the 10-class classification task. However, the performance on certain low-frequency subtypes was markedly lower than that of competing models. This phenomenon may stem from the significant underrepresentation of low-frequency subtypes in the dataset. Such class imbalance compels comparative models to adopt more conservative

prediction strategies, whereby samples are classified as low-frequency subtypes only when the prediction confidence exceeds a predetermined threshold.

To further address application scenarios that require a focus on specific genomic segments—such as vaccine design—we developed an alternative subtyping workflow. This workflow was evaluated on a 10-subtype classification task. As shown in Fig. 3c, Infl-BERT outperformed all competing models in the subtype classification task (Table 2), achieving the highest F1-scores across nine subtypes. Notably, for the H1N2 subtype, its F1-score exceeded that of the second-best model, DNABERT2, by 22% (Supplementary Table S11), demonstrating a pronounced advantage. It is worth noting that when the analysis was restricted to the HA and NA gene segments, DNABERT2 and NT also exhibited strong performance, while other models remained moderately competitive, which highlights its effectiveness in scenarios where subtype identification depends primarily on HA and NA sequences.

In summary, for the scenario without distinguishing specific genomic segments, these experiments found that directly applying the cross-species pre-trained DNABERT-2 model in its original form yielded poor performance in recognizing low-frequency subtypes, and traditional machine learning models also underperformed on low-frequency subtype identification. However, Infl-BERT (the modified DNABERT-2 model), after optimization using the two-stage training strategy proposed in this study, achieved improvements in identifying both common and low-frequency subtypes, and exhibited consistent improvements across both five-class and 10-class tasks. This demonstrates that Infl-BERT retains a relative advantage in low-frequency subtype identification even as the task scale expands. Given that the HA and NA segments are key determinants of IAV subtype combinations, the overall model performance in this experiment was relatively satisfactory, with Infl-BERT outperforming other methods



**Figure 3** Experimental results of subtyping identification tasks. (a) Performance evaluation across five common subtypes without distinguishing specific genomic segments; (b) performance evaluation across 10 subtypes without distinguishing specific genomic segments; (c) performance evaluation across 10 subtypes using only the HA and NA segments.

**Table 3** Infl-BERT was compared with baseline models on respiratory virus recognition.

|                               | Method     | ACC          | Precision    | Recall       | F1-score     |
|-------------------------------|------------|--------------|--------------|--------------|--------------|
| Respiratory virus recognition | DNABERT2   | 0.997        | 0.997        | 0.997        | 0.997        |
|                               | MegaDNA    | 0.996        | 0.996        | 0.996        | 0.996        |
|                               | NT         | 0.999        | 0.999        | 0.999        | 0.999        |
|                               | LR         | 0.999        | 0.999        | 0.999        | 0.999        |
|                               | RF         | 0.985        | 0.985        | 0.985        | 0.985        |
|                               | SVM        | 0.978        | 0.978        | 0.978        | 0.978        |
|                               | Influ_BERT | <b>0.999</b> | <b>0.999</b> | <b>0.999</b> | <b>0.999</b> |

and achieving the highest identification performance as measured by the F1-score.

### Performance evaluation for extended downstream tasks

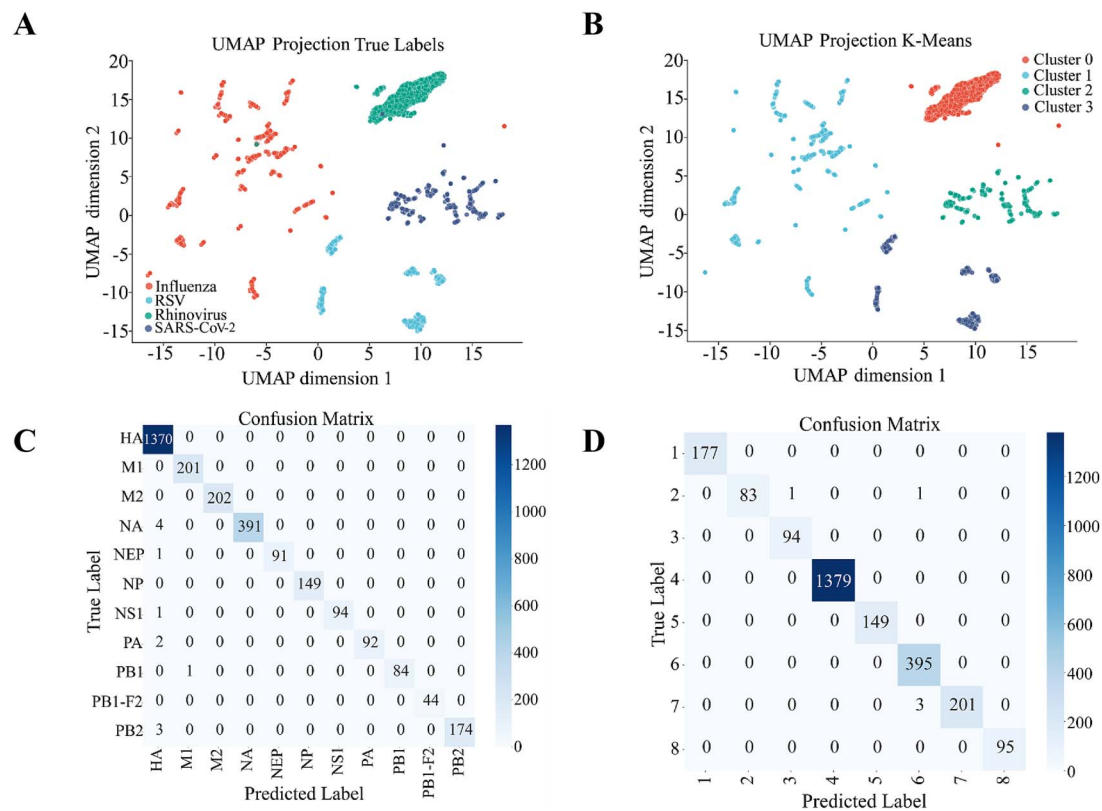
To further validate the performance of Infl-BERT, we extended its application to three additional downstream tasks: identification of common respiratory viruses, identification of IAV genomic segments and functional genes, and prediction of IAV pathogenicity.

In the first task, leveraging the domain-adapted pretrained model, we identified four common respiratory viruses—SARS-CoV-2, rhinovirus, RSV, and influenza virus. As summarized in Table 3, Infl-BERT achieved superior performance compared to other baseline models. We further visualized the embeddings derived from Infl-BERT (Fig. 4a and b), which exhibited clear spatial separation between virus types. Applying the K-means clustering algorithm to these embeddings revealed well-defined cluster structures, confirming the model's ability to learn discriminative sequence representations.

In the second task, we applied Infl-BERT to the identification of influenza virus genomic segments and functional genes. As shown in

Fig. 4c, the functional gene identification results indicate that while the HA gene exhibited relatively more prediction errors compared to other genes, Infl-BERT still achieved excellent overall performance on this task. In Fig. 4d, Infl-BERT accurately classified nearly all genomic segments, with only two misclassifications occurring in Segment 3 and Segment 6, underscoring its robust capability for segment-level identification.

In the third task, we addressed the prediction of IAV pathogenicity. Given the limited availability of labeled pathogenicity data in real-world scenarios, this task emphasized evaluating the model's ability to learn from small datasets. Specifically, we designed two experiments: one with 200-samples and another with 400-samples. As summarized in Table 4, in the 200-samples experiment, Infl-BERT achieved an F1 score of 78.9%, outperforming the second-best model (RF) by 2.7% and excelling across all evaluation metrics. With increased data in the 400-samples setting, Infl-BERT's F1 score further improved to 82%, ranking second, yet performing nearly on par with the top-ranked RF model. Notably, DNABERT-2 exhibited poor performance under the 200-sample setting, whereas Infl-BERT achieved an F1-score ~10% higher. These results demonstrate that Infl-BERT possesses strong representation learning capabilities and generalizes effectively when labeled data is scarce.



**Figure 4** Experimental results of extended tasks. (a and b) Embeddings of diverse respiratory virus sequences, extracted using Influa-BERT, were visualized to examine their distribution patterns; these embeddings were subsequently clustered using the K-means algorithm; (c) confusion matrix for functional gene prediction; (d) confusion matrix for gene segment prediction.

**Table 4** Influenza BERT was compared with baseline models on pathogenicity prediction.

|             | Method     | ACC          | Precision    | Recall       | F1-score     |
|-------------|------------|--------------|--------------|--------------|--------------|
| 200-samples | DNABERT2   | 0.693        | 0.703        | 0.693        | 0.692        |
|             | MegaDNA    | 0.579        | 0.604        | 0.579        | 0.570        |
|             | NT         | 0.647        | 0.665        | 0.647        | 0.645        |
|             | LR         | 0.755        | 0.768        | 0.755        | 0.755        |
|             | RF         | 0.762        | 0.767        | 0.762        | 0.762        |
|             | SVM        | 0.693        | 0.692        | 0.693        | 0.691        |
|             | Influ_BERT | <b>0.790</b> | <b>0.802</b> | <b>0.790</b> | <b>0.789</b> |
| 400-samples | DNABERT2   | 0.818        | 0.822        | 0.818        | 0.819        |
|             | MegaDNA    | 0.779        | 0.778        | 0.779        | 0.779        |
|             | NT         | 0.815        | 0.814        | 0.815        | 0.814        |
|             | LR         | 0.806        | 0.826        | 0.806        | 0.806        |
|             | RF         | <b>0.823</b> | <b>0.834</b> | <b>0.823</b> | <b>0.824</b> |
|             | SVM        | 0.783        | 0.783        | 0.783        | 0.783        |
|             | Influ_BERT | 0.820        | 0.827        | 0.820        | 0.821        |

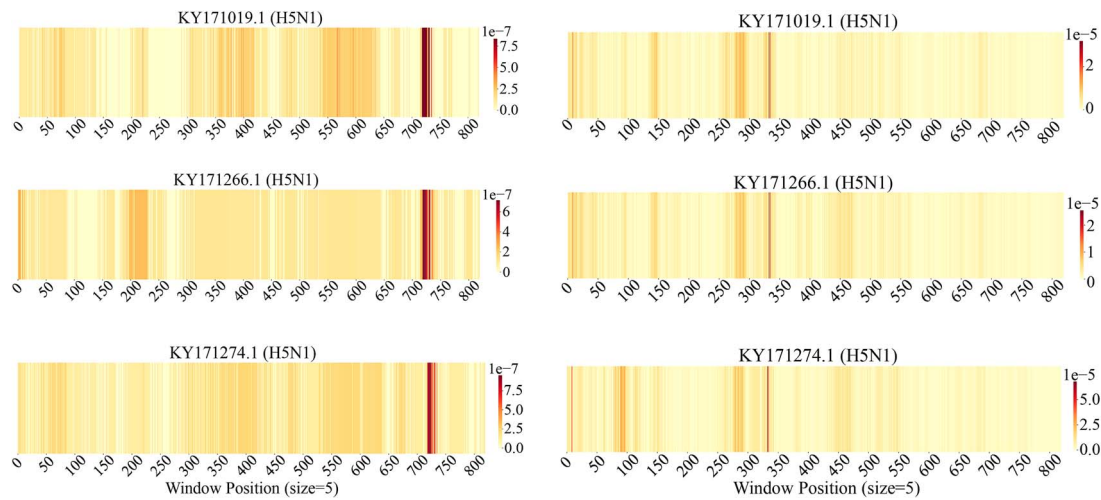
200-samples: the experiment comprised 100 high-pathogenicity and 100 low-pathogenicity samples. The bold values indicate the best performance relative to the other models.

Interpretability analysis by perturbation

To investigate how the model captures informative features within influenza sequences, we designed a masking-based sliding window perturbation analysis. By systematically masking input sequences and observing variations in the model’s predictions, we reveal differences in feature attention patterns between DNABERT-2 and Influa-BERT (Fig. 5), providing preliminary insights into the latter’s improved performance in low-frequency subtype identification. Using three H5N1 influenza virus NS segments as representative examples (Fig. 5; left:

DNABERT2, right: Influa-BERT), Influa-BERT exhibited highly consistent attention patterns across all sequences. Notably, the model focused on the 274–276 nt region, corresponding to codon 92 of the NS1 protein—a known virulence-determining site where substitution of glutamic acid (E) confers cytokine resistance in H5N1 strains [38]. Additionally, the 330–340 nt region identified by the model maps to the multifunctional central domain of NS1, which promotes viral translation through interaction with eIF4GI [39]. These observations suggest that Influa-BERT can capture biologically meaningful sequence features even in the absence of explicit functional annotations.





**Figure 5** Illustration of perturbation analysis for three selected H5N1 sequences. The left panel shows the results of DNABERT-2, while the right panel corresponds to Influenza-BERT.

In contrast, DNABERT-2 exhibited widespread attention across the sequences, with multiple regions marked as important; however, the concentrated attention to the 720–730 nt region lacks reported functional annotation. A similar pattern was observed in the M segment of the H3N2 subtype (Supplementary Fig. S2), where DNABERT-2 displayed numerous attention regions without focusing on specific functional sites. In contrast, Influenza-BERT showed specific attention to nucleotides 38–41 nt, which correspond to positions 50–53 nt of M1 mRNA after accounting for the ~12 nt host-derived 5' leader sequence. This region encodes amino acid 9 of the M2 protein and overlaps the 5' splice donor site of the mRNA, thus serving dual functional roles under strong evolutionary constraint [40]. These findings suggest that Influenza-BERT may leverage such key sequence features to achieve precise genomic localization in complex influenza subtype recognition tasks.

## Conclusion and discussion

Although masked pre-trained models such as DNABERT-2, trained on cross-species datasets, and generative models like MegaDNA have demonstrated excellent performance across various biological sequence analysis tasks, they exhibit notable limitations in capturing knowledge representations specific to influenza virus sequences. These constraints stem primarily from the highly long-tailed distribution of influenza virus data, relatively short sequence lengths, and frequent genetic mutations.

To address these challenges, we proposed a domain-adaptive pretrained model, Influenza-BERT. In the subtype identification task, Influenza-BERT demonstrated robust effectiveness even without explicitly distinguishing genomic segments, achieving substantial improvements in both F1-score and recall, particularly for low-frequency subtypes. When trained exclusively on the HA and NA segments, Influenza-BERT still outperformed other models. To investigate whether the observed performance gains originate from improved sequence representations, we visualized the embeddings directly obtained from the pre-trained checkpoints of Influenza-BERT and DNABERT-2 using UMAP (Supplementary Fig. S3). The resulting projections reveal that Influenza-BERT constructs a more discriminative feature space, particularly for low-frequency subtypes. In contrast, DNABERT-2 exhibits substantial mixing among these low-frequency subtypes,

indicating that its general-purpose genomic pre-training fails to capture features that effectively distinguish low-frequency variants, instead mapping them to closely overlapping regions in the embedding space. By comparison, Influenza-BERT successfully decouples these low-frequency subtypes, providing clearer separation. Collectively, these results demonstrate that domain-specific pre-training yields more informative and discriminative sequence representations. Furthermore, we extended its application to tasks including common respiratory virus classification, IAV gene fragment and functional gene identification, and IAV pathogenicity prediction. Across all these tasks, Influenza-BERT consistently achieved superior performance, underscoring its powerful representational capacity for influenza genomic sequences.

From a biological perspective, IAV subtype determination is primarily governed by the HA and NA segments. Nevertheless, AI-based approaches can accurately identify subtypes without explicit segments for two main reasons. First, co-evolutionary relationships among viral gene segments allow discriminative information related to HA and NA to be implicitly captured in other segments [41]. This segment-agnostic identification strategy is particularly advantageous in rapid detection scenarios or when limited sequencing depth leads to missing HA and NA segments. In contrast, for applications that require high accuracy or target specific genomic segments, such as vaccine design, a segment-specific approach is more suitable. This workflow can also be integrated with segment identification models to enable the rapid retrieval of HA and NA segments from large-scale sequence datasets, thereby facilitating subsequent subtype identification processes. Second, the model is capable of learning statistical patterns from sequence data and mapping inputs directly to subtype labels.

Despite the encouraging progress achieved by Influenza-BERT in influenza virus sequence recognition, some limitations remain. First, the current model relies on supervised closed-set learning during fine-tuning, which limits its ability to recognize novel subtypes absent from the training data. Future work will focus on addressing the challenge of identifying emerging influenza subtypes, approached from two complementary perspectives. First, for known subtypes, we aim to enhance the model's closed-set learning capability. Specifically, Influenza-BERT will be employed to learn representations of HA and NA

segments separately, enabling any known Hx segment to be accurately mapped to its corresponding HA label and any Nx segment to its corresponding NA label. These segment-level predictions can then be combined to generate the final subtype classification, allowing comprehensive identification of all known subtype combinations. Second, for entirely novel subtypes, we propose a confidence-based anomaly detection strategy integrated with embedding-space outlier analysis. Samples with model confidence scores below a defined threshold, and those identified as outliers in the embedding space, will be flagged to prioritize candidates for subsequent experimental validation. This approach aims to simultaneously ensure accurate recognition of known subtypes while providing early-warning capabilities for novel, previously uncharacterized subtypes.

### Key Points

- We developed an influenza-specific corpus and proposed Infl-BERT using a two-stage pretraining strategy, significantly enhances the model's capability to identify low-frequency subtypes and overcomes the limitations of general-purpose models when dealing with long-tailed data distributions, short sequence lengths, and high mutation rates.
- We further applied Infl-BERT to three downstream tasks: common respiratory virus identification, influenza A virus gene fragment and functional gene identification, and IAV pathogenicity prediction. Experimental results demonstrate that Infl-BERT achieves consistently superior performance almost all tasks, highlighting its powerful representational capacity.
- We introduced a sliding-window perturbation analysis to enhance model interpretability. This method enables the visualization of subtype-discriminative regions and uncovering key attention patterns essential for accurate influenza subtype identification.

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### Supplementary material

Supplementary material is available at *Briefings in Bioinformatics* online.

### Conflicts of interest

The authors declare no conflicts of interest related to the content of this article.

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### Data availability

All genome sequences were downloaded from <https://github.com/0000111/Influenza-BERT/tree/main>.

### Code availability

Source code is written in PyTorch and available at <https://github.com/0000111/Influenza-BERT>, [https://huggingface.co/rongye1/Influenza\\_BERT](https://huggingface.co/rongye1/Influenza_BERT) and <https://ngdc.cnbc.ac.cn/biocode/tool/BT008097> under the MIT license.

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